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W4 Case Study: Face Recognition

Thursday, October 15, 2026

READINGS Burton, "Face Perception" (2024); Tsao, "Face Values" (2019); Chang & Tsao, "The Code for Facial Identity in the Primate Brain" (2017), excerpts

CONTENTS

1 Abstraction, without and within . . . 1
2 Face perception . . . 1
3 Large-scale Neural Organization . . . 2
4 The theory: 50D code for faces . . . 3
5 **Representation and computation** . . . 7
6 Marrian Levels . . . 8

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1 Abstraction, without and within

- **Abstraction in the science of cognition**
  - **A simple model.** "Our results demonstrate that at least one subdivision of IT cortex can be explained by an explicit, simple model, and 'black box' explanations are unnecessary." (Chang & Tsao 2017, p. 1024)
- **Abstraction as a tool of cognition.**
  - **Abstract computation.** "The finding proved that face cells are not encoding the identities of specific individuals in the IT cortex. Instead they are performing an axis projection, a much more abstract computation." (Tsao 2019, p. 13)

2 Face perception

Computational-level constraints

- **Short-term, view-bound 2D image as input.**
- **Enduring, view-invariant output.**

- **The invariance problem.** "Recognizing an object despite these visual transformations is called the invariance problem, and it became clear to us that a major function of the face-patch network is to overcome this impediment." (Tsao 2019, p. 7)

Computational-level goals

- **Face categorization.** (Categorizing some objects as faces.)
- **Face recognition.** (Identifying individuals by their faces.)
- **Emotion recognition.** (Identifying affective state by face.)
- **Other target properties:** age, gender, aesthetic valence.
- **Achieve these goals via...**
  - **Face analysis:** structured description of face and facial features.

Debates

- **Are face areas really special for faces?**
  - "This experiment revealed an astonishing result: almost every single cell in the area identified through fMRI was dedicated to processing faces." (Tsao 2019, p. 3)
- **Are there "Jennifer Aniston cells" in the face areas?**
  - **The expectation.** "I had expected the face area would contain cells that responded selectively to specific individuals, analogous to orientation-selective cells in the primary visual cortex that each respond to a specific edge orientation. . . . Contrary to my expectation, each cell seemed to fire vigorously for almost any face." (Tsao 2019, p. 4)
- **Is the face recognition system exemplar based?**

3 Large-scale Neural Organization

Sequenced architecture

- **Six patches.** ". . . it became clear that there are in fact six face patches in each of the brain's two hemispheres (making a dozen golden eggs total)." (Tsao 2019, p. 5)

- **An assembly line.** "This observation of a stereotyped pattern suggested that the patches might constitute a kind of assembly line for processing faces." (Tsao 2019, p. 5)
- **Interconnected.** "We discovered that whenever we stimulated one face patch, the other patches would light up, but the surrounding cortex would not, indicating that, indeed, the face patches are strongly interconnected." (Tsao 2019, p. 5)

Six face patches

(1) **PL.** Posterior lateral.
(2) **ML.** Middle lateral. *View specific. (up vs. down / left vs. right)*
(3) **MF.** Middle fundus. *View specific.*
(4) **AL.** Anterior lateral. *Partial view-invariance (up or down / left or right).*
(5) **AF.** Anterior fundus.
(6) **AM.** Anterior medial. *View-invariant. Final output stage of IT face processing.*

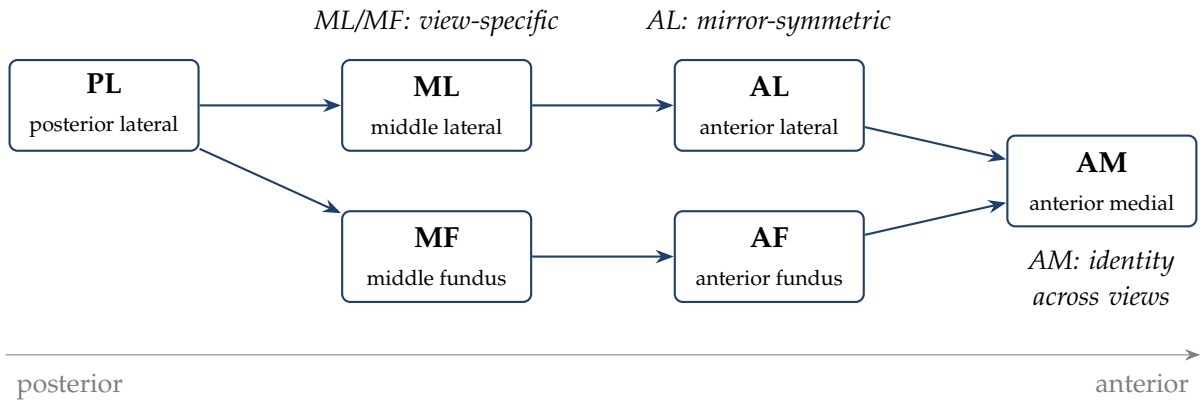


Figure 1. Main flow through the face patches, from the back of the temporal lobe to the front. Top row: lateral patches; bottom row: patches in the fundus (depth) of the superior temporal sulcus. Stage labels from Tsao (2019, p. 7).

4 The theory: 50D code for faces

The idea

- **Shape**
  - **... skeletal form:** "Think of the shape features as those defined by the skeleton—how wide the head is or the distance between the eyes." (Tsao 2019, p. 9)

- **Appearance**
  - **... as surface form:** "The appearance features define the surface texture of the face (complexion, eye or hair color, and so on)." (Tsao 2019, p. 9)
- **50 dimensions**
  - **Face space.** "By taking the top 25 principal components for shape and the top 25 for appearance, we created a 50-dimensional face space." (Tsao 2019, p. 9)
- **Face vectors**
  - **A point in face space.** Each face corresponds to a location in the 50D face-space. This is called a "vector", because it specifies the direction and distance of the point from the origin.
  - **The origin.** "... the average face (i.e., the face at the origin of the face space) ..." (Chang & Tsao 2017, p. 1016)
- **Face patches as linear projectors**
  - **Spike-triggered average (STA).** "To quantify neuronal tuning to the 50 dimensions of the face space, responses of each neuron were first used to calculate a 'spike-triggered average' (STA) stimulus ..., i.e., the average stimulus that triggered the neuron to fire ..." (Chang & Tsao 2017, p. 1015)
  - **Dot product.** "If AM cells are in fact linearly combining different features, then geometrically what an AM cell is doing is simply taking a dot product between an incoming face and a specific direction in face space defined by the cell's STA ..." (Chang & Tsao 2017, p. 1020)
  - **A shadow.** "... a mathematical operation known as the dot product, whose geometric representation is the projection of a vector onto an axis (like the sun projecting the shadow of a flagpole onto the ground)." (Tsao 2019, p. 12)

Evidence 1: variation along dimensions

- **The experiment.** "In our experiment, we randomly drew 2,000 faces and presented them to a monkey while recording cells from two face patches." (Tsao 2019, p. 9)
- **Ramps.** "We found that almost every cell showed graded responses—resembling a ramp slanting up or down—to a subset of the 50 features ..." (Tsao 2019, p. 9)
- **The equation.** "response of face cells = weight matrix × 50 face features" (Tsao 2019, p. 12)

- **Shape vs. appearance.** "We found that most ML/MF cells showed stronger tuning to shape dimensions than to appearance dimensions, while AM cells showed the opposite trend . . . " (Chang & Tsao 2017, p. 1015)
- **Same scheme.** "Thus, the fundamental difference between ML/MF and AM lies in the axes being encoded (shape versus shape-free appearance), not in the coding scheme." (Chang & Tsao 2017, p. 1020)

Challenge: Axis theory vs. Exemplar theory

- **The challenge.** Are cells encoding feature axes, or are cells encoding face exemplars?
  - "Many models of object recognition assume an exemplar-based representation . . . in which object recognition is mediated by units tuned to exemplars of specific objects that need to be recognized." (Chang & Tsao 2017, p. 1019)
- **Evidence for exemplar theory.** "More direct support for an exemplar-based model comes from recordings in face patch AM, where a subset of cells have been found to respond extremely sparsely to only a few identities, invariant to head orientation . . . " (Chang & Tsao 2017, p. 1019)
- **The crucible**
  - "To decide whether a cell is coding an exemplar or an axis, the critical question is, what is the shape of tuning along axes in the plane orthogonal to the STA axis?" (Chang & Tsao 2017, p. 1020)
  - "If this plane constitutes a 'null space' in which all faces elicit the same response, this would be indicative of axis coding. Alternatively, if there is Gaussian tuning along axes within this plane, this would be indicative of exemplar coding." (Chang & Tsao 2017, p. 1020)

Evidence 2: variation orthogonal to dimensions

- **Flat tuning.** "Surprisingly, the tuning of AM neurons was largely flat along orthogonal axes and showed no clear bias for Gaussian nonlinearity." (Chang & Tsao 2017, p. 1020)
- **Sparse cells too.** "However, when we looked at tuning of these sparse neurons in a 2-d space spanned by the STA and an orthogonal axis, they showed a drastic nonlinearity along the STA but nearly no tuning along the orthogonal axis . . . " (Chang & Tsao 2017, p. 1020)

- **Metamers:** faces that differ only along axes orthogonal to a neuron's preferred axis, and so do not affect its response. "... these faces are 'metameric' because they differ by a large vector lying in the cell's null space ..." (Chang & Tsao 2017, p. 1020)

Evidence 3: reconstructing faces from neural signals

- **Inverting the equation.** "50 face features = (1/weight matrix) × response of face cells" (Tsao 2019, p. 12)
- **Reconstruction.** "By combining the predicted feature values across all 50 dimensions, we could reconstruct the face that the monkey saw." (Chang & Tsao 2017, p. 1017)
- **Almost indistinguishable.** "Astonishingly, the prediction turned out to be almost indistinguishable from the actual face." (Tsao 2019, p. 12)
- **About 200 cells.** "We show that it is possible to decode any human face using just ~200 face cells from patches ML/MF and AM, once faces are defined in the proper space (as vectors in a 'shape-appearance' space)." (Chang & Tsao 2017, p. 1024)

Big picture implication

- **No identity detectors in IT.**
  - **Surprising.** "This is surprising because it means there are no detectors for identities of specific individuals in the face patch system—even though intuitively one might expect this, especially after observing a sparse AM cell ..." (Chang & Tsao 2017, p. 1024)
  - **Not in IT.** "It would seem then that the Jennifer Aniston cells do not exist, at least not in the IT cortex." (Tsao 2019, p. 13)
- **IT generates efficient face descriptions.**
  - **About 50 parameters.** "Facial recognition in the IT cortex thus rests on a set of about 50 numbers in total that represent the measurement of a face along a set of axes." (Tsao 2019, p. 13)

Advantages of Axis theory over Exemplar theory (axis metric vs. distance metric)

- **Efficiency.** "First, axes are more efficient, allowing a smaller number of units to achieve similar performance." (Chang & Tsao 2017, p. 1024)

- **Robustness.** "Linearly pooling inputs with similar signal but independent noise would help improve the signal quality . . . , leading an axis model to perform better in noisy conditions." (Chang & Tsao 2017, p. 1024)
- **Flexibility.** "Finally, an axis metric endows downstream areas reading out the activity of AM with greater flexibility to discriminate along a variety of different dimensions." (Chang & Tsao 2017, p. 1024)
- **In sum.** "In sum, axis coding is more flexible, efficient, and robust to noise for representation of objects in a high-dimensional space compared to exemplar coding." (Chang & Tsao 2017, p. 1024)

5 Representation and computation

Is this research about representation or computation?
Computation: transition from patch to patch

- **Wiring.** "How are cells wired to transform an object, defined in pixel coordinates, into a shape-appearance feature vector representation?" (Chang & Tsao 2017, p. 1026)
- **A deep network.** "Most likely, in the brain, such a representation is produced by an architecture similar to a hierarchical feedforward deep-network." (Chang & Tsao 2017, p. 1026)

Representation: dimensional analysis

- **Format.** "Regardless of how the wiring is accomplished by the brain, we believe our insight that the brain is formatting faces into a shape/appearance feature vector representation is important in itself . . . " (Chang & Tsao 2017, p. 1026)

Linear projectors

- **Projection machines.** "So far, all of our results point to a model of face cells as linear projection machines. While simple, this model is also surprising because it means face cells are performing a rather abstract mathematical computation." (Chang & Tsao 2017, p. 1022)
- **Is this the right diagnosis?**
  - There is ambiguity here about whether the AM cells are *performing the computation*, or whether they are *registering the output of the computation*.

- That is, is their primary goal *computation*, or is their primary goal *presenting a representation*?
- Alternative diagnosis: the whole face-patch system is the linear projector. The AM cells are registering the output of the process.

Questions:

- What decodes the 50D representation? how is it decoded?
- Isn't the case of human faces ecologically invalid for monkeys?

6 Marrian Levels

Computational level:

- **Measuring faces.** "Regardless of the details of how non-face objects are represented in IT, we believe the computational goal of object representation is likely the same across all of IT including face patches. The present study suggests that within face patches, this goal is to set up a coordinate system to measure faces rather than to explicitly identify faces, since almost every single face cell we recorded was susceptible to face metamers." (Chang & Tsao 2017, p. 1027)

Algorithmic level:

- **Linear projection.** "Furthermore, we reveal the mechanism underlying this remarkably efficient code for facial identity: linear projection of incoming face vectors onto a specific axis (the 'STA' axis)." (Chang & Tsao 2017, p. 1024)

Neural level:

- The face-patch network.
- But what about at a neuron-by-neuron level?